Becoming a Self-Regulated Learner : Implications for Web-Based Education

(Dembo, Junge, & Lynch. Chapter11)

Introduction

- **Web-based learning or training**
  - Any learning that uses Web-based content or communication via the Internet focusing on flexibility and the demands of individual learners (Kjeldsen, Krogsdal, & Gomme, 2003).

- **Distance learning**
  - May be defined as learning that occurs in a different place than teaching, utilizing special aspects of course design, instructional methods, and delivery modalities.

Introduction

- **U.S. Department of Education**

- Distance learning has traditionally faced higher student dropout rates (between 10%-50%) than classroom-based courses. The situation with Web-based distance learning, with non-completions in some online corporate university courses as high as 70%-80%.

Three categories of factors that predict dropout in distance learning：

① **Student factors 學生因素**

- Educational preparation (教育準備)
- Motivational (動機)
- Persistence attributes (持久性)

② **Situational factors 情境因素**

- Family support (家庭支持)
- Changes in life (生活變化)

③ **Educational system factors 教育制度因素**

- Quality and difficulty level of instruction.
- Materials quality and availability of instructional feedback.

Introduction

- Initiative and self-discipline are two important learner characteristics that are needed for success in distance learning.

- Garrison discussed the limitations of autonomy and control to fully understand learner behavior, and pointed out that the psychological or cognitive aspects of the learning process have been generally ignored in the conceptualization and adoption of self-directed learning in distance education.

SELF-REGULATION
自我調制

SELF-REGULATION

- **Self-regulation (自我調制)**
  - Defined as the ability of learners to control the factors or conditions affecting their learning.

- Learner‘s self-regulatory beliefs and processes are highly correlated with academic achievement.

- The student must incorporate a combination of cognitive, metacognitive, motivational and behavioral processes to attain the highest possible level of academic achievement.

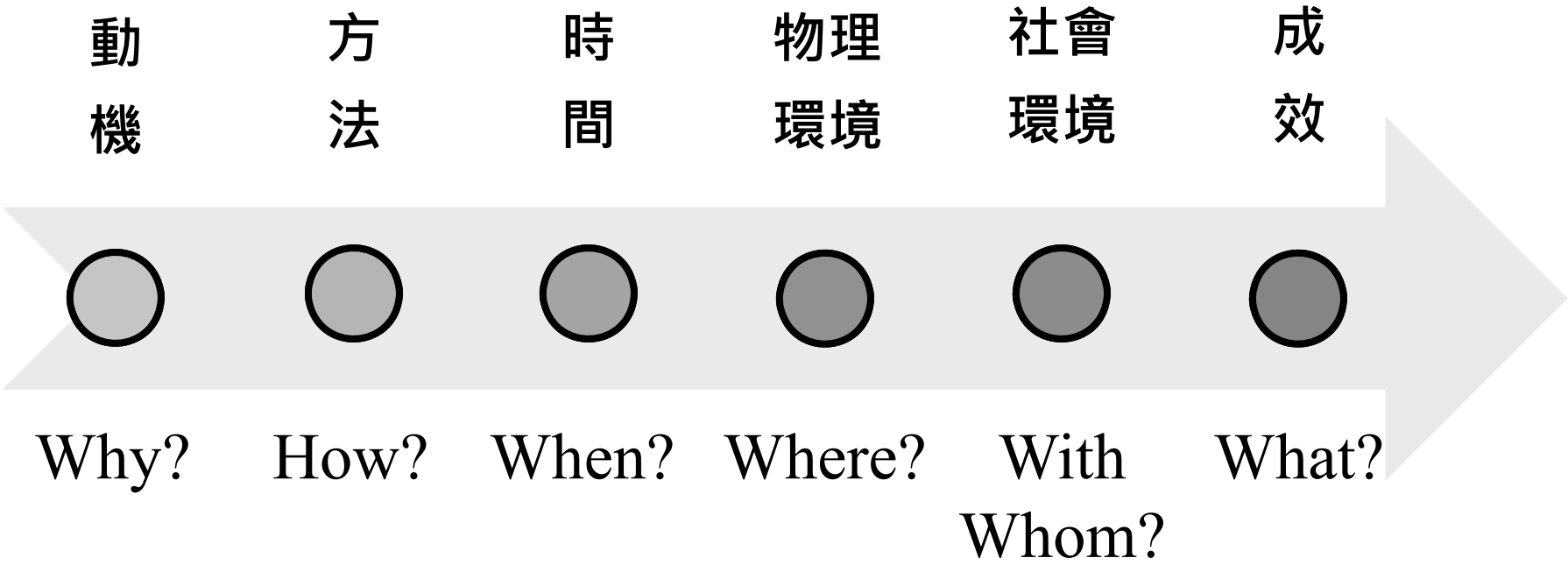
Components of Self-regulation
自我調制的要素

Components of Self-regulation

- Zimmerman (1994, 1998) presented a conceptual framework of self-regulation that consists of <u>six underlying psychological dimensions</u> that learners can self-regulate by employing specific processes.

  - Motives
  - Methods of learning
  - Use of time
  - Physical Environment
  - Social Environment
  - Performance

Components of Self-regulation



Motive 動機

Motive 動機

- Why learners choose to learn, that is, their motivation for learning.

- Student characteristic influencing success in distance learning (Hofmann, 2003; Qureshi, Morton, & Antosz, 2002).

Motive 動機

- Schunk, Pintrich, & Meece (2008) have identified three observable behaviors that can be used as indicators of motivation :

1. Active choice

- What tasks learners decide to engage in.

2. Persistence

- The willingness of learners to maintain behavior when tasks are difficult, challenging, or boring.

3. Effort

- How hard learners work on a task.

Motive 動機

❖ **Goal 目標**

- High achievers report using goal setting more frequently and more consistently than do low achievers.
- Learning goals increase learning if learners believe that :

  (a) The goals are possible to achieve with the time and resources available.

  (b) They have, or will acquire, the knowledge to achieve the goal.

  (c) The goals are specific, explicit, and difficult.

Motive 動機

- **“Three C” goals**

- When learners face extraordinarily challenging tasks with a minimum of training and a great deal of motivation, “three C” goals (Concrete、Current 、Challenging) may influence their learning.

Method 方法

Method 方法

- How learners self-regulate learning.

- The key underlying process regulating method is the use of learning strategies.

- Higher achieving students use more learning strategies than do lower achieving students (Zimmerman & Martinez-Pons, 1988).

Method 方法

- Weinstein and Mayer － three different categories of learning strategies that learners must acquire.

① Rehearsal strategies（複誦）：

- Can be effective for learning factual material.
- Copying, reciting words or definitions are all examples of rehearsal strategies.
- Their limitation, however, is that they make few connections between new information and the knowledge that is already in long-term memory.

Method 方法

② <u>Elaboration strategies（精緻化）</u>：

- Helps retention by linking new information with knowledge already in long-term memory.
- These strategies are useful for improving the recall of names, categories, sequences, or groups of items.

③ <u>Organizational strategies（組織）</u>：

- These strategies help learners remember information by allowing them to structure academic content.
- Students can learn to structure content through visual representations like diagrams, matrices, sequences, and hierarchies (Dembo, 2004).

Time 時間

Time 時間

- When and for how long to study.

- In considering the causes of online learner dropouts, Cookson (2000) noted the flexiability to manage learning time as a primary factor.

Time 時間

- Tuckman (2002) studied procrastination(拖延) in a Web-based course with many performance deadlines.

- He found that more serious procrastinators were more likely to utilize rationalizations, less likely to self-regulate, and earned lower grades.

Time 時間

- Marron reported that in a poll taken by a career Web site, greater than 90% of the 1,244 employees sampled surfed the Internet for Information irrelevant to their jobs at work.

- Lavoie and Pychyl found that over half of their 208 subjects reported frequent Internet procrastination, which was positively correlated with trait procrastination and negative emotions.

Time 時間

- Self-regulated learners know how to manage their time because they are aware of deadlines, how long it will take to complete each assignment and their own learning processes.

- They prioritize learning tasks, evaluating the more difficult from easier tasks in terms of the time required to complete them. They are aware of the need to evaluate how their study time is spent and to reprioritize as necessary.

Physical Environment
物理環境

Physical Environment 物理環境

- Where learners learn and which Instructional supports they employ.

- Zimmerman and Martinez-Pons (1986) found that high achievers reported greater use of environmental restructuring than did low-achieving students.

Physical Environment 物理環境

- In terms of physical space, Web-based learners generally have the option of accessing their courses via computers at home or elsewhere.

- Environmental selection and restructuring：
  - Ensuring the study environment is conductive to learning and restructuring or changing it as necessary.

Social Environment
社會環境

Social Environment 社會環境

- Social environment refers to with whom the learner studies (Zimmerman, 1998).

- Active learners who are motivated to achieve tend to seek help when it is needed.

- They know when help is needed, make the decision to seek help, identify sources of help, use appropriate strategies to obtain the help, and finally, they evaluate the help received (AJeven et al., 2003).

Social Environment 社會環境

- Hill (2001) presented several strategies and techniques:

  - Instructors and course designers can employ to develop a sense of community in Web-based learning environments.

  - Lessens the social isolation of distance study and fosters shared learning activities, including mutual helping behaviors (Rovai, 2002).

Social Environment 社會環境

- An emotionally "safe," comfortable online environment enables learners to communicate openly without fear of negative criticism.

- It is also important to encouragement for learners to maintain regular contact with one another and with the instructor throughout the course.

- Learners should also be provided with multiple means of accessing and sharing information.

Performance 成效

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- Performance addresses questions relating specifically to overt behavior, or what learners do in pursuit of their learning goals.

- Self-regulated learners are aware of the learning outcomes they expect (have set for themselves), are sensitive to not having achieved those outcomes (if such is the case), and are able to adjust their behavior accordingly in order to make up for any behavioral deficiency in attaining their learning goals.

Performance 成效

- Learners can be taught to regulate their learning performance through setting learning goals, monitoring learning progress, and evaluating learning.

- Learners can be taught to develop their own daily progress logs and self-assessment checklists keyed to their learning goals and to employ them regularly during the monitoring and evaluation phases of their learning.

- These metacognitive strategies enhances development of learner autonomy and life-long learning skills (Clark & Shatkin, 2003; Kassop, 2003).

SUMMARY

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- Although an increasing number of students are enrolling in Web-based courses, dropout rates remain problematic.

- Many authors have identified the importance of autonomy, initiative, and self-discipline in web-based learning.

- Six important components of self-regulatory behavior and related motivation and learning strategies derived from classroom-based research have been identified and applied to Web-based learning.

SUMMARY

- This model can be used both for identifying possible Web-based learner characteristics that can limit successful learning (i.e., prevention) and as a guide to developing intervention strategies to deal with learning problems.
- Educators realize that low achieving students may lack important self-regulatory skills that must be taught either before enrolling in a Web-based course or while taking courses.
- These skills can and must be directly taught to learners and acquired by them through guided practice.